Boammani Aser **Lompo**

Msc

About me

ML Research Scientist (MILA/ETS) with expertise in Multimodal Reasoning, Foundation Models, and AI Safety. Strong theoretical foundation (ENS Paris, IMO) combined with end-to-end research delivery: NeurIPS workshop, ICLR submission, and open-source benchmarks.

Webpage

https://aser97.github.io/Blog/

Areas of specialization

LLM • VLM • RL • AI Safety • Maths

Interests

- Photography & Content Creation
- **Competitive Chess (2000 Elo)**

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@AserLompo

Aser Lompo

Aser97

Research & Work Experience

2021 **Internship, Department of Statistics, CREST, ENSAI**
Intern · Rennes
Research on unimodal bandits for ranking systems. Keywords: Recommender Systems, Multi-armed Bandits, Learning to Rank. Supervisor: Romaric Gaudel

2020 **Internship in a Research Laboratory in Cryptology, INRIA**
Intern · Paris
Measuring privacy of cryptographic systems and studying the properties of optimal-privacy systems. Supervisor: Catuscia Palamidessi

Degrees

2025 **MSc Machine Learning Research**
MILA / ETS Montréal · Ecole Technologie Superieure (Montreal)

2023 **MSc Mathematics, Vision & Learning**
MVA · Ecole Normale Superieure Paris-Saclay

2021 **MSc Fundamental Mathematics**
Agrégation · Ecole Normale Superieure (Rennes)

Tech Stack

Languages: Python • C++ • JavaScript • Bash
Frameworks: PyTorch • Transformers • JAX • Hugging Face • vLLM • LoRA/PEFT
Tools: Docker • Weights & Biases • Paperspace • Git • MCP
Focus Areas: Multimodal Reasoning • Foundation Models • RL • AI Safety • Maths

Portfolio

Safety Guardrail
- Fine-tuned **Qwen2.5-7B** + LoRA
- 9-signal youth distress taxonomy
- **F1 = 0.85** Github

Visual-TableQA
- **2.5k** complex table images, **9k** QA pairs
- Cost-efficient pipeline (**<$100**), **+17%** fine-tuning gain
- Github

Number Encoding
- LESA + Xval encoder upgrade for LLMs
- **+12%** on numerical reasoning Github

More on my webpage.

Certificates & Grants

2024 Recipient of R-SADC scholarship 2024 (**23,000$**)
2018–22 French Government Excellence Grant (**50,000$**)
2024 Best Oral Presentation **Acfas 2024** (91e Congrès)
2014 Honourable Mention at the International Mathematics Olympiade (**IMO**) in Cape Town 2014
2015 **Gold medal** at the Panafrican Mathematics Olympiade in Abuja
2015 Winner of the National Math Contest in Burkina Faso 2015

Publications

2026 B.A. Lompo. *Impact of Spurious Correlation on Gradient Descent Dynamics and Generalization.* (submission soon). [GitHub]
2025 B.A. Lompo, M. Haraoui. *MODALITY-SWAP DISTILLATION: Rendering Textual Reasoning into Visual Supervision.* Under review at **ICLR 2026**.
2025 B.A. Lompo, M. Haraoui. *Visual-TableQA: Open-Domain Benchmark for Reasoning over Table Images.* **NeurIPS 2025** Workshop. [PDF]
2024 B.A. Lompo, T-D. Le, P. Jouvet, R. Noumeir. *Multi-objective Representation for Numbers in Clinical Narratives.* **CoRR 2024**. [PDF]
2021 C-S. Gauthier, R. Gaudel, E. Fromont, B.A. Lompo. *Parametric Graph for Unimodal Ranking Bandit.* **ICML 2021**. [PDF]

Reviewer: ICCV 2023 • CVPR 2026 • ICLR 2026

Languages

English	Fluent (C2)
French	Native (C2)

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